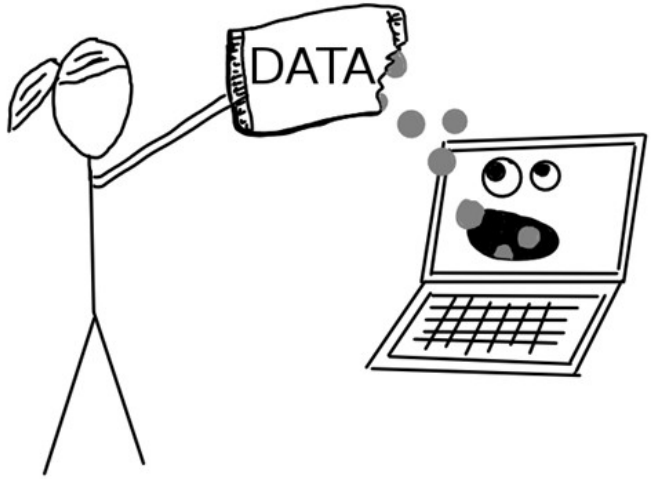

ENSA Oujda- Cycle Ingéneur

Ingénierie Data Sciences et Cloud Computing

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Deep Learning
Churn_Modelling Project

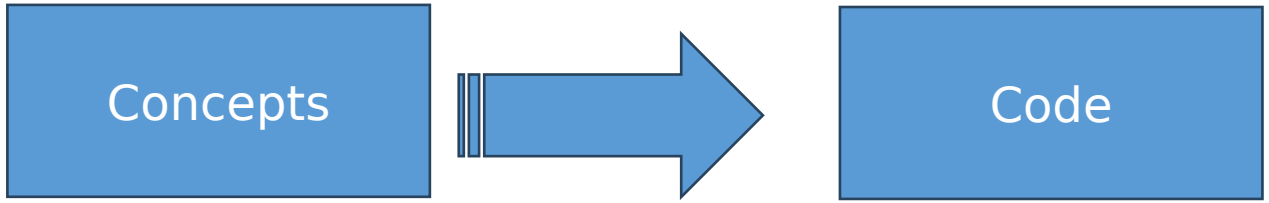
Abdelwahab Naji - Enseignant chercheur
Abdelwahab.naji@gmail.com
ENSAO, Université Mohamed premier - Oujda
https://www.youtube.com/c/AbdelwahabNaji

how to learn machine learning well

If you can’t implement it, then you don’t understand it

How to practice

- Recall: this is a hands-on course
- Get your hands dirty!



- **Remember: a real job / project involves actually writing code**

Churn_Modelling Project

1. Data Preparation
2. Model
3. Prediction
4. Evaluation
5. Improve Performance

1. Data Preparation

1. Load Data set in a Dataframe
2. Visualize Data with DataFrame
3. Identify the type of problem
4. identify non-predictive features without any analysis
5. Identfiy features type
6. Extract columns, lines from a DataFrame
7. Encode categorical variables
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10. Separate features & target
11. Split data
12. Rescale Data

1. Data Preparation

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5. Extract data from the dataframe

```
import numpy as np

import matplotlib.pyplot as plt

import pandas as pnd

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import LabelEncoder, OneHotEncoder

from sklearn.preprocessing import StandardScaler
```

```
dataset
```

```
dataset=pnd.read_csv("Churn_Modelling.csv")
```

	RowNumber	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
0	1	15634602	Hargrave	619	France	Female	42	2	0.00	1	1	1	101348.88	1
1	2	15647311	Hill	608	Spain	Female	41	1	83807.86	1	0	1	112542.58	0
2	3	15619304	Onio	502	France	Female	42	8	159660.80	3	1	0	113931.57	1
3	4	15701354	Boni	699	France	Female	39	1	0.00	2	0	0	93826.63	0
4	5	15737888	Mitchell	850	Spain	Female	43	2	125510.82	1	1	1	79084.10	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
9995	9996	15606229	Obijiaku	771	France	Male	39	5	0.00	2	1	0	96270.64	0

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1	2	15647311	Hill	608	Spain	Female	41	1	83807.86	1	0	1	112542.58	0
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...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
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Binary type: Binary classification problem

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Non predictive features

	RowNumber	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
0	1	15634602	Hargrave	619	France	Female	42	2	0.00	1	1	1	101348.88	1
1	2	15647311	Hill	608	Spain	Female	41	1	83807.86	1	0	1	112542.58	0
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3	4	15701354	Boni	699	France	Female	39	1	0.00	2	0	0	93826.63	0
4	5	15737888	Mitchell	850	Spain	Female	43	2	125510.82	1	1	1	79084.10	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
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5. **Extract data from the dataframe**

```
dataset1=dataset.iloc[:,3:14]
```

```
dataset1
```

	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
0	619	France	Female	42	2	0.00	1	1	1	101348.88	1
1	608	Spain	Female	41	1	83807.86	1	0	1	112542.58	0
2	502	France	Female	42	8	159660.80	3	1	0	113931.57	1
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4	850	Spain	Female	43	2	125510.82	1	1	1	79084.10	0
...	...	...	...	...	...	...	...	...	...	...	...
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7. Encode categorical variables
8. Encode binary variable
9. Encode ordinal variables

categorical

binary

	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
0	619	France	Female	42	2	0.00	1	1	1	101348.88	1
1	608	Spain	Female	41	1	83807.86	1	0	1	112542.58	0
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3	699	France	Female	39	1	0.00	2	0	0	93826.63	0
4	850	Spain	Female	43	2	125510.82	1	1	1	79084.10	0

continuous

discret

continuous

1. Data Preparation

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9. Encode ordinal variables

```
onehotencoder=OneHotEncoder()

pays=onehotencoder.fit_transform(dataset[['Geogr
aphy’]])

pays=pays.toarray()

result1=pnd.DataFrame(pays,columns=['pays1','pa
ys2','pays3’])

result2=result.drop('pays3', axis=1)
```

result1

	pays1	pays2	pays3
0	1.0	0.0	0.0
1	0.0	0.0	1.0
2	1.0	0.0	0.0
3	1.0	0.0	0.0
4	0.0	0.0	1.0

result2

	pays1	pays2
0	1.0	0.0
1	0.0	0.0
2	1.0	0.0
3	1.0	0.0
4	0.0	0.0

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```
onehotencoder=OneHotEncoder()

pays=onehotencoder.fit_transform(dataset[['Geogr
aphy']])

pays=pays.toarray()

result1=pnd.DataFrame(pays,columns=['pays1','pa
ys2','pays3'])

result2=result.drop('pays3', axis=1)

result3=dataset1.drop('Geography', axis=1)
```

result2

	pays1	pays2
0	1.0	0.0
1	0.0	0.0
2	1.0	0.0
3	1.0	0.0
4	0.0	0.0

```
dataset1
```

	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
0	619	France	Female	42	2	0.00	1	1	1	101348.88	1
1	608	Spain	Female	41	1	83807.86	1	0	1	112542.58	0
2	502	France	Female	42	8	159660.80	3	1	0	113931.57	1
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4	850	Spain	Female	43	2	125510.82	1	1	1	79084.10	0
...	...	...	...	...	...	...	...	...	...	...	...
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1. **Load Data set in a Dataframe**
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8. Encode binary variable
9. Encode ordinal variables

result2

	pays1	pays2
0	1.0	0.0
1	0.0	0.0
2	1.0	0.0
3	1.0	0.0
4	0.0	0.0

```
result3
```

	CreditScore	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
0	619	Female	42	2	0.00	1	1	1	101348.88	1
1	608	Female	41	1	83807.86	1	0	1	112542.58	0
2	502	Female	42	8	159660.80	3	1	0	113931.57	1
3	699	Female	39	1	0.00	2	0	0	93826.63	0
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```
dataset2=pnd.concat([result2,result3],axis=1)
```

```
dataset2
```

	pays1	pays2	CreditScore	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
0	1.0	0.0	619	Female	42	2	0.00	1	1	1	101348.88	1
1	0.0	0.0	608	Female	41	1	83807.86	1	0	1	112542.58	0
2	1.0	0.0	502	Female	42	8	159660.80	3	1	0	113931.57	1
3	1.0	0.0	699	Female	39	1	0.00	2	0	0	93826.63	0
4	0.0	0.0	850	Female	43	2	125510.82	1	1	1	79084.10	0

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7. **Encode categorical variables**
8. **Encode binary variable**
9. Encode ordinal variables

```
label_encoder=LabelEncoder()
dataset2['Gender']=
label_encoder.fit_transform(dataset2['Gender'])
```

```
dataset2
```

	pays1	pays2	CreditScore	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
0	1.0	0.0	619	Female	42	2	0.00	1	1	1	101348.88	1
1	0.0	0.0	608	Female	41	1	83807.86	1	0	1	112542.58	0
2	1.0	0.0	502	Female	42	8	159660.80	3	1	0	113931.57	1
3	1.0	0.0	699	Female	39	1	0.00	2	0	0	93826.63	0
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Dataset not contain an ordinal variable

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7. **Encode categorical variables**
8. **Encode binary variable**
9. **Encode ordinal variables**
10. **Separate features & target**

```
X=dataset2.iloc[:,:11]
y=dataset.iloc[:,11]
```

11. Split data
12.

	pays1	pays2	CreditScore	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary
0	1.0	0.0	619	0	42	2	0.00	1	1	1	101348.88
1	0.0	0.0	608	0	41	1	83807.86	1	0	1	112542.58
2	1.0	0.0	502	0	42	8	159660.80	3	1	0	113931.57
3	1.0	0.0	699	0	39	1	0.00	2	0	0	93826.63
4	0.0	0.0	850	0	43	2	125510.82	1	1	1	79084.10

```
0    1
1    1
2    0
3    0
4    1
     ..
```

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12. Rescale Data

```
X_train,X_test,y_train,y_test=train_test_split(X,y,tes
t_size=0.25)
```

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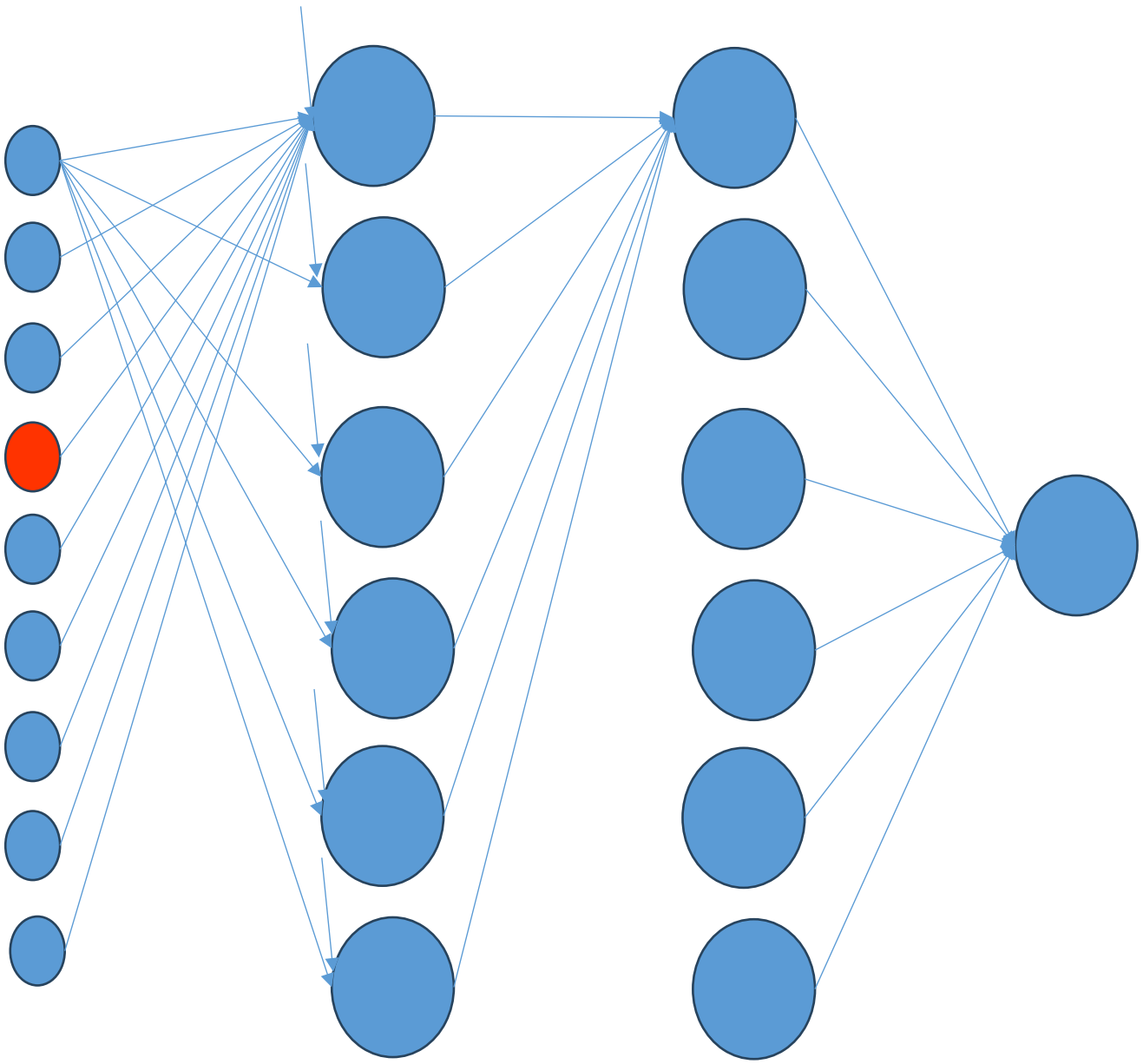
```
sc=StandardScaler()

X_train[:,[2,4,5,6,7,10]]=sc.fit_transform(X_train[:,[2,4,5,6,7,10]])

X_test[:,[2,4,5,6,7,10]]=sc.transform(X_test[:,[2,4,5,6,7,10]])
```

	pays1	pays2	CreditScore	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary
0	1.0	0.0	619	0	42	2	0.00	1	1	1	101348.88
1	0.0	0.0	608	0	41	1	83807.86	1	0	1	112542.58
2	1.0	0.0	502	0	42	8	159660.80	3	1	0	113931.57
3	1.0	0.0	699	0	39	1	0.00	2	0	0	93826.63
4	0.0	0.0	850	0	43	2	125510.82	1	1	1	79084.10

2. Model

1. Choose the technique (Linear Regression, **Neural Netwrok**, SVR,…)
2. choose an architecture
3. Create a neural network according to the architecture
   - Initialize the model
   - create input layer and hidden layers
   - Create output layer
4. Compile
   - optimizer
   - Loss
   - metrics



3. Prediction

4. Evaluation

5. Improve Performance